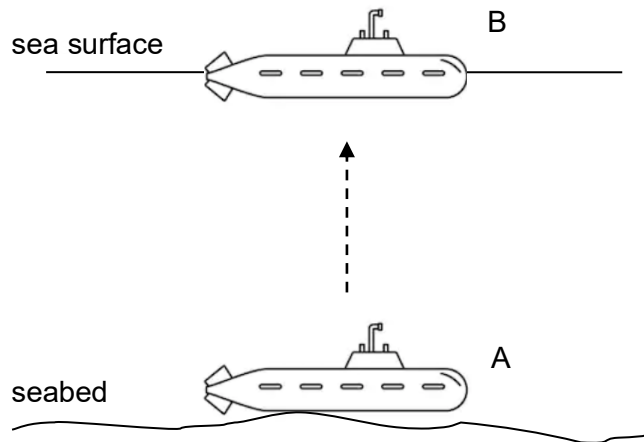


- 7 A submarine is initially at rest on the seabed as shown in A in the diagram below. It then rises and floats on the sea surface as shown in B.



If the total weight of the submarine and its contents is  $W$  and the upthrust on the submarine is  $U$ , by comparing the two positions A and B, which of the following is correct?

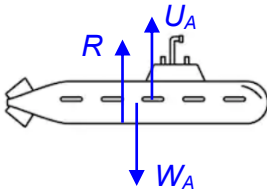
**A**  $U$  has increased and  $W$  has decreased.

- B**  $U$  has decreased and  $W$  remains the same.  
**C** Both  $U$  and  $W$  have decreased.  
**D** Both  $U$  and  $W$  have increased.

**Ans: C**

Since the submarine is fully submerged in A and partially submerged in B, the weight of fluid displaced is lower in B,  $U$  has decreased.

At A (since the submarine is resting on the seabed),

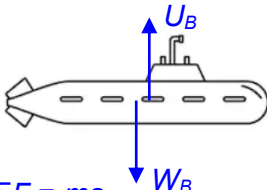


$$\Sigma F = ma$$

$$U_A + R - W_A = 0$$

$$W_A = U_A + R$$

At B,



$$\Sigma F = ma$$

$$U_B - W_B = 0$$

$$W_B = U_B$$

Since  $U_B < U_A$ ,  $W_B < W_A$